

Pooja Kiran

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EXPERIENCE

Independent AI Security Researcher & Engineer

Aug. 2024 – Present

Self-Directed Research

Tempe, AZ

- Built and evaluated security controls across agent runtime enforcement, AWS IAM, model supply-chain security, LLM red teaming, training-data integrity, and adversarial ML across **19 repositories**, with threat models, reproducible tests, and evidence-backed claims.
- Standardized verification with GitHub Actions, SARIF/Code Scanning, CodeQL, Bandit, Trivy, dependency/secret scanning, Docker, Elastic Security, Prometheus, and runbooks documenting controls, failure modes, and limitations.

Business & Compliance Lead — AEROSSEC

Aug. 2025 – Dec. 2025

Honeywell Aerospace Technologies × Arizona State University

Tempe, AZ

- Led business/compliance analysis for a proposed airline PSS protection layer and built a **\$120K first-year commercialization scenario** with a **five-year financial model** presented to ASU and Honeywell stakeholders.

Graduate Teaching Assistant — IT Grader

Jan. 2025 – Oct. 2025

Ira A. Fulton Schools of Engineering, Arizona State University

Mesa, AZ

- Evaluated approximately **85 undergraduate web-development submissions per semester** and **52 graduate cybersecurity assignments** covering policy, compliance, risk analysis, and security controls.

SELECTED AI SECURITY PROJECTS

MCP Agent Security Gateway | *Python, FastAPI, MCP/JSON-RPC 2.0, Elastic Security, Docker*

Jul. 2026 – Present

- Engineered an inline tool-call gateway with default-deny authorization, capability validation, **55 prompt-injection patterns**, PII/exfiltration signals, anti-SSRF egress controls, rate limiting, and tamper-evident audit evidence.
- Validated **718 automated tests at 82.46% statement coverage** across Python 3.10–3.12 at the cited snapshot; implemented **9 Elastic Security rules and 21 core SIEM tests**.

AWS Agent Identity Guard | *Python, AWS IAM, SARIF 2.1.0, GitHub Code Scanning*

Aug. 2026 – Sep. 2026

- Built a static IAM analyzer with **25 deterministic rules** covering wildcards, `iam:PassRole`, `sts:AssumeRole`, privilege escalation, weak trust policies, audit tampering, and missing permission boundaries.
- Verified **235 passing tests with 3 credential-gated skips**; integrated text, JSON, and SARIF CI enforcement, with a 500-policy synthetic benchmark measuring **1.146 ms p95 and 1,846 policies/sec**.

HF Model Provenance Scanner | *Python, SafeTensors, GGUF, ONNX, Keras, SARIF*

Jul. 2026 – Sep. 2026

- Developed a non-executing model supply-chain scanner using pickle-opcode inspection, AST/taint analysis, dependency/provenance checks, obfuscation detection, and multi-format binary inspection.
- Verified **241 passing tests at 75.81% statement coverage**; detected **33/33 committed adversarial fixtures** with **0 actionable findings across 4 committed benign samples**.

Dataset Poisoning Detector | *Python, FastAPI, Kafka, Redis, Prometheus, scikit-learn*

Jul. 2026 – Present

- Built a training-data integrity pipeline combining statistical, Isolation Forest, ensemble, spectral, label-aware, drift, deduplication, quarantine, streaming, and fail-closed readiness controls.
- Verified **199 passing tests at 90.88% statement coverage** with a **90% CI gate**; label-flip screening measured **0.55 / 0.60 / 0.70 F1 at 5% / 10% / 20% poisoning** in the committed benchmark.

EDUCATION

Arizona State University

Tempe, AZ

Master of Science in Information Technology (Security) – GPA: 3.87/4.00

Aug. 2024 – May 2026

M. S. Ramaiah University of Applied Sciences

Bengaluru, India

Bachelor of Technology in Computer Science and Engineering – CGPA: 8.44/10

Aug. 2019 – Aug. 2023

TECHNICAL SKILLS

AI Security: MCP/JSON-RPC, Tool Security, Prompt Injection, LLM Red Teaming, Model Supply Chain, Training-Data Integrity, Adversarial ML

Cloud/AppSec: AWS IAM, Least Privilege, Trust Policies, Permission Boundaries, Threat Modeling, API Security, Capability Authorization

Engineering: Python, Rust, C++, FastAPI, pytest, Hypothesis, GitHub Actions, SARIF, CodeQL, Bandit, Trivy, Docker, Elastic Security, Prometheus

Certifications: AWS Academy Cloud Architecting; AWS Academy Cloud Security Foundations